DIGITAL MIRROR - Problem Statement

DIGITAL MIRROR is an AI-powered multimodal behavioral analysis and simulation platform designed to identify how modern social media algorithms gradually influence human psychology, emotional stability, attention span, and digital dependency. Modern social media platforms maximize user engagement using algorithm-driven recommendation systems, short-form videos, emotionally triggering content, and addictive visual patterns. Continuous exposure to such content may gradually affect human behavior, emotional regulation, cognitive focus, sleep patterns, and communication habits. Existing digital wellbeing systems mainly focus on screen-time tracking and application blocking, but they fail to computationally analyze the deeper psychological and behavioral impact caused by algorithmic content exposure. The proposed system aims to bridge this gap by analyzing user interactions across text, audio, image, and video formats to identify behavioral patterns such as dopamine dependency, emotional instability, attention fragmentation, cognitive fatigue, and algorithmic influence. DIGITAL MIRROR introduces a research-oriented behavioral modeling approach using computational psychological indicators such as Dopamine Trigger Score, Attention Fragmentation Index, Sleep Disruption Score, and Digital Behavioral Risk Index (DBRI). The system generates behavioral insights, emotional influence analysis, addiction indicators, and future behavioral simulations to help users understand how digital platforms may gradually reshape human behavior.